

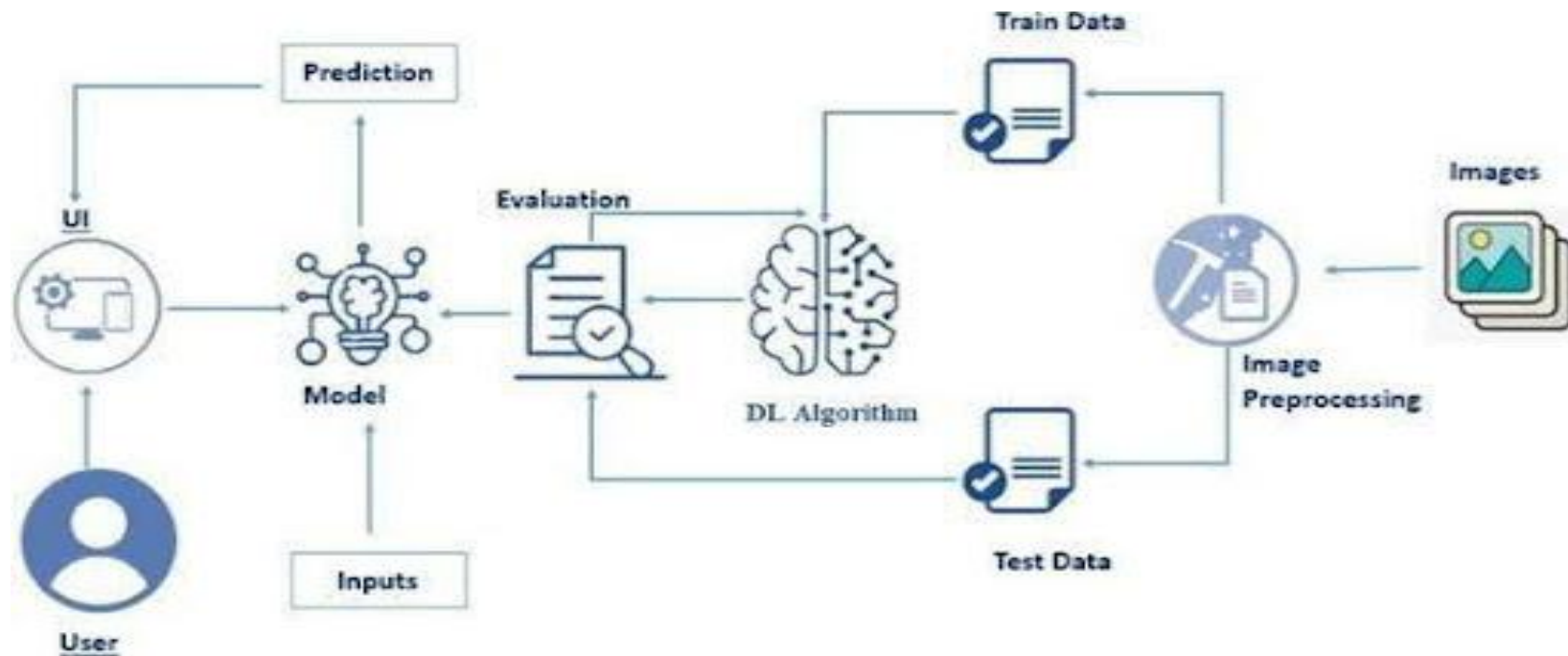
Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	LTVIP2026TMIDS81120
Project Name	Smart Sorting: Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of how information moves within the fruit freshness classification system. A neat and well-structured DFD graphically illustrates the system requirements by showing how image data enters the application, how it is processed by the machine learning model, and how prediction results are generated and delivered to the user..

Example:



Fruit Freshness Classification using Machine Learning is an important application that aims to automatically identify whether fruits are fresh or spoiled with high accuracy. By analyzing image data and visual features such as color, texture, and surface patterns through deep learning algorithms, this project enables reliable quality assessment without manual inspection. The goal is to provide instant and accurate predictions to consumers, vendors, and quality inspectors, helping reduce food waste, improve safety, and maintain consistent product standards in real-world environments.

User Stories – Flood Prediction System

a.Data Collection & Upload

US-01:

As a Data Analyst, I want to upload fruit image datasets so that the system can use labeled images for training the classification model.

US-02:

As a System Administrator, I want the system to validate uploaded image datasets so that corrupted or unsupported files are rejected.

b. Data Preprocessing

US-03:

As a Data Analyst, I want the system to preprocess images by resizing and normalizing them so that the model receives consistent input.

US-04:

As a Data Analyst, I want the dataset to be automatically split into training and testing sets so that model performance can be evaluated properly.

c. Model Training & Algorithm

US-05:

As a Machine Learning Engineer, I want to train a convolutional neural network using the prepared dataset so that the system learns visual patterns of fresh and spoiled fruits.

US-06:

As a Machine Learning Engineer, I want to tune training parameters and apply data augmentation so that prediction accuracy improves and overfitting is reduced.

d. Model Evaluation

US-07:

As a System Analyst, I want to evaluate the trained model using metrics such as accuracy and validation loss so that I can measure its effectiveness.

US-08:

As a Project Stakeholder, I want to compare training results and model performance so that the most reliable model can be selected.

e. User Interface & Prediction

US-09:

As a User (Consumer/Vendor/Inspector), I want to upload a fruit image through the web interface so that I can check its freshness.

US-10:

As a User, I want to receive a clear prediction result (Fresh / Spoiled) so that I can make safe and informed decisions.

US-11:

As a User, I want the system to provide fast real-time predictions so that I can quickly assess fruit quality